PROJECT DOCUMENTATION

CAR RENTAL BOOKING SYSTEM

A PHP & MySQL web application for managing vehicle rentals and reservations

Student's Name	Ismail Hossain
Trainee ID	1297568
Batch	PWAD/USSAL-M/01/73
Project Name	Car Rental Booking System

Prepared for project presentation

Development approach: standard PHP forms using GET/POST and full page reloads. No AJAX or Fetch API.

1. Project Highlights

This project is designed to represent the main activities of a small-to-medium car rental business. Customers can find vehicles and request bookings, while administrators and staff manage vehicles, payments and handovers.

Feature	What it does
Customer accounts	Registration, login, profile and booking history.
Car catalogue	Browse cars with photos, specifications, category and rental price.
Booking management	Choose pickup/return dates and locations; submit a booking request.
Availability checking	Prevent conflicting reservations for the same car and dates.
Admin dashboard	Manage cars, customers, bookings, categories and payment records.
Staff operations	Handle vehicle pickup, return, inspection and damage reports.
Reports and records	View booking status, payment details and rental summaries.
No-AJAX implementation	Use regular HTML forms, PHP GET/POST processing and page reloads.

Main project outcome

A role-based rental website with a clear booking process, database-backed records and basic business administration. Payment gateway integration and other advanced services can be added in a later version.

2. Introduction, Objectives and Users

Introduction

The Car Rental Booking System is a web-based application that helps a rental company manage its cars and customer reservations in one place. It reduces manual record keeping and makes booking information easier to track.

Project objectives

- Make it easier for customers to view available cars and request rentals.
- Store customer, vehicle, booking and payment information in MySQL.
- Help administrators manage day-to-day rental operations.
- Check booking date conflicts before accepting a reservation.
- Provide a simple, understandable interface using Bootstrap.

Main user roles

Customer	Register, browse cars, request bookings, view own bookings and payment records.
Admin	Manage users, cars, bookings, categories, payments and reports.
Staff / Manager	Manage assigned pickups, returns, vehicle condition and handover information.

Project scope

The system covers vehicle listing, customer accounts, booking requests, booking status, payment records and vehicle handover/return. Actual online payment processing is a future enhancement unless a payment gateway is separately integrated.

3. System Features and Modules

Customer module

- Create an account, log in and update profile information.
- Browse cars and view photos, seats, fuel type, transmission and daily price.
- Submit pickup date, return date, pickup location and rental requirements.
- View booking status, booking history and payment information.
- Cancel a request according to the rental business rules.

Admin module

- Dashboard summary for cars, customers, bookings and payment records.
- Add, edit, view and remove cars; manage categories and brands.
- Approve, reject or cancel booking requests.
- Manage customer accounts, staff and driver assignments.
- Record payments, issue invoice details and review business reports.

Staff / manager module

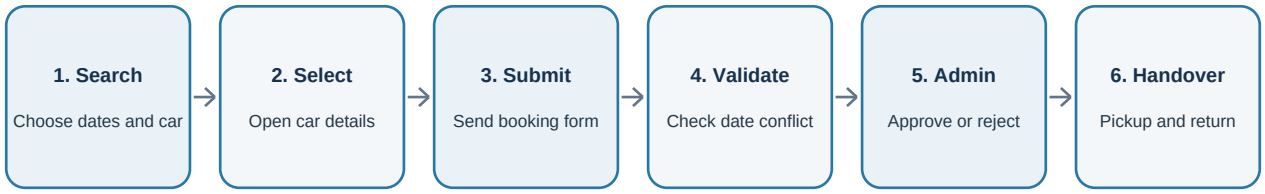
- View assigned bookings and customer details needed for handover.
- Record pickup and return time, fuel level and vehicle condition.
- Report damage or late return and update the handover status.

Booking status values

Pending • Confirmed • Rejected • Cancelled • Ongoing • Completed • Overdue

4. Booking Workflow (Easy-to-Understand Diagram)

The diagram shows how a customer booking moves through the system, from searching for a car to completing the rental.



Workflow explanation

- Customer logs in and searches cars by rental dates and other preferences.
- Customer opens a car details page and submits the booking form.
- PHP checks required fields, date order and overlapping bookings.
- The system calculates the rental amount and saves the request as Pending.
- Admin reviews the request and confirms or rejects it.
- Staff records vehicle pickup and return; the booking becomes Completed after inspection.

Important availability rule

A car must not be booked by two customers for overlapping confirmed or ongoing rental periods. The system must compare the requested pickup/return dates with existing bookings for that car. A simple Available/Not Available flag alone is not enough.

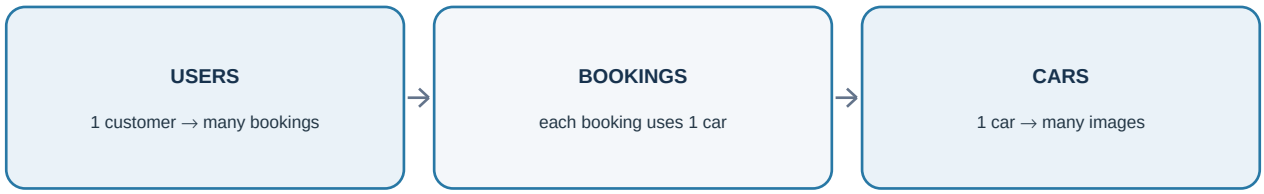
5. Database Design and Relationship Diagram

Database name: **car_rental_db**. MySQL stores the system's users, cars, bookings and supporting records.

Core tables

Table	Purpose
users	Customer, admin and staff account details
roles	Role names and access types
cars	Car name, category, brand, price and specifications
categories / brands	Vehicle grouping and manufacturer information
car_images	Images linked to each car
bookings	Customer, car, dates, location, total and status
payments	Payment amount, method, date and verification status
drivers / driver_assignments	Driver details and booking assignments
services / booking_services	Optional rental add-ons
reviews / wishlists	Customer feedback and saved cars
maintenance / car_damage_reports	Vehicle maintenance and condition records

Simplified relationship diagram



Other links: CATEGORIES → CARS; BRANDS → CARS; BOOKINGS → PAYMENTS; DRIVERS → DRIVER_ASSIGNMENTS → BOOKINGS.

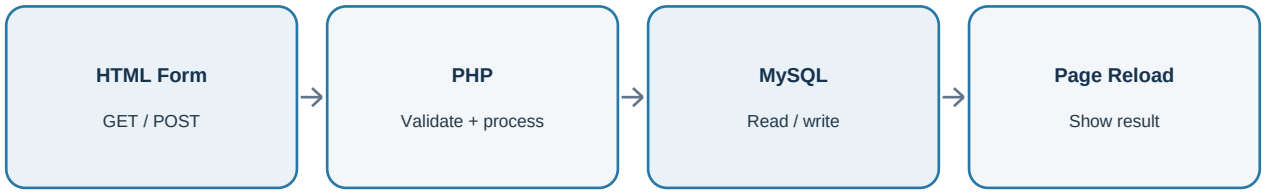
6. Technology Stack and Application Structure

Technology stack

Technology	Purpose
HTML5 / CSS3	Page structure and styling
Bootstrap 5	Responsive layout, navigation, forms and tables
JavaScript	Basic client-side interaction and form assistance
PHP 8	Server-side processing, validation and session handling
MySQL	Database storage and queries
Apache / XAMPP	Local development server

No-AJAX approach

This project will not use AJAX or the Fetch API. Forms will submit through normal HTTP GET/POST requests. PHP will process the request on the server, save or retrieve database information, and then load the appropriate page again.



Suggested folder structure

```
car_rental/
■■■ admin/   ■■■ customer/   ■■■ staff/
■■■ config/dbconfig.php
■■■ includes/ (header, footer, auth)
■■■ assets/ (css, js, images)
■■■ uploads/ (cars, profiles)
■■■ index.php, cars.php, booking.php, login.php, register.php
```

7. Security, Testing and Conclusion

Security practices

- Use password_hash() to store passwords and password_verify() to check them.
- Use prepared statements for database queries.
- Use PHP sessions to protect login-required pages.
- Check user roles before allowing admin or staff actions.
- Validate all important form data on the server.
- Escape displayed user input with htmlspecialchars().
- Validate uploaded file type and size; do not trust file names.
- Check booking date conflicts before saving or confirming a booking.

Local setup and testing

- Install XAMPP and start Apache and MySQL.
- Create the car_rental_db database in phpMyAdmin.
- Import the project's SQL table structure.
- Place the project folder inside htdocs.
- Update config/dbconfig.php with the local database settings.
- Open the project through http://localhost/car_rental/ and test each user role.

Future enhancements

Online payment gateway integration, email/SMS notifications, map-based pickup locations, advanced analytics and mobile application support can be added in future versions.

Conclusion

The Car Rental Booking System provides a structured way to manage rental cars, customers, reservations and vehicle handovers. It uses PHP and MySQL with standard form submissions, keeping the implementation suitable for an intermediate web development project and presentation.

Thank You